

UNITED INSTITUTE OF MANAGEMENT

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BCA MINOR / MAJOR PROJECT SYNOPSIS

on

TIMETABLE & STUDY PLANNER GENERATOR

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1. Introduction

1.1 Background & Motivation

Time management is one of the biggest challenges a student faces during their academic life. With multiple subjects, assignments, and a fixed number of free hours in a week, planning an effective study routine becomes difficult. Traditionally, students prepare their study timetable manually on paper or in a notebook, which is time-consuming, hard to update, and often abandoned after a few days.

A simple digital tool that automatically builds a study schedule, while still letting the student adjust it by hand, can make weekly planning far easier. This is the idea behind the "Timetable & Study Planner Generator".

1.2 Problem Statement

- Manual timetables take time to create and are difficult to update once subjects or hours change.
- Subjects are often unevenly distributed, with some days overloaded and others left empty.
- There is no easy way to rearrange a paper timetable once it is written down.
- Students have no simple, dedicated tool that does this planning automatically and is still easy to edit.

1.3 Objectives of the Project

- To let a student add subjects and set the weekly study hours needed for each one.
- To automatically generate a balanced weekly timetable using a simple, fixed scheduling rule (no AI required).
- To allow the student to fine-tune the generated timetable using a drag-and-drop editor.
- To store subject and timetable data safely in a MySQL database.
- To provide a simple, clean interface that any student can use without training.

1.4 Key Features & Benefits

- **Structured Input:** Students add subjects along with the exact hours needed per week.
- **Automatic Generation:** A round-robin scheduling rule spreads every subject evenly across the week.
- **Drag-and-Drop Editor:** Any subject block can be dragged into a different day or time slot.
- **Persistent Storage:** The timetable is saved so it can be viewed or edited again later.

Overall, the tool saves planning time, spreads subjects fairly across the week, and still leaves the student in full control of the final plan.

1.5 Scope & Significance of the Project

Beyond individual use, this project can later be extended for coaching institutes or schools to manage timetables for a whole batch, and enhanced with reminders, progress tracking, and a mobile app.

2. Technologies Used

Front-End

HTML5, CSS3, Bootstrap, and JavaScript (for the drag-and-drop timetable editor and form validation).

Back-End

PHP - handles subject data, the round-robin scheduling logic, and communication with the database.

Database

MySQL - stores student accounts, subjects, and the generated timetable.

Tools & Software

Item	Details
Code Editor	VS Code
Local Server	XAMPP / WAMP Server (PHP & MySQL)
Browser	Google Chrome (for testing)
Database Tool	phpMyAdmin

Hardware Requirements

- Processor: Intel Core i5 or higher
- RAM: 4GB minimum (8GB recommended)
- Storage: 10GB free disk space
- Operating System: Windows 10/11, Linux, or macOS

3. Layout of Proposed Work

1. Home / Login Page

The landing page lets a student register or log in. Navigation includes Home, My Subjects, Timetable, and Logout.

2. Subject Management Module

Students can add a subject with the hours needed per week, view the list of added subjects, and remove a subject if it is no longer needed.

3. Timetable Generation Module

On clicking "Generate Timetable", the system applies a round-robin rule on the server side (PHP) that spreads each subject across the selected days and hours, so no subject is clustered on a single day.

4. Drag-and-Drop Editor Module

The generated timetable is shown as a weekly grid. Using JavaScript, the student can drag any subject block and drop it onto another slot; the system swaps the two slots instantly and saves the change.

5. Database Module

All subject details and the final timetable are stored in MySQL against the student's account, so the plan can be reopened and edited at any time.

6. System Architecture

The system follows a three-tier model:

- **Frontend:** HTML, CSS, Bootstrap, JavaScript (responsive interface).
- **Backend:** PHP for scheduling logic and request handling.
- **Database:** MySQL for storing subjects and the timetable.

Student Modules

- Student Registration
- Student Login
- Add / View / Delete Subjects
- Generate Weekly Timetable
- Edit Timetable (Drag-and-Drop)
- Logout

Admin Modules

- Admin Dashboard

- Manage Student Accounts
- View All Generated Timetables
- Change Password

4. Role of Team Member

Member Name	Role of Team Member
Komal Singh	Front-end Development & Drag-and-Drop Editor (HTML5, CSS3, Bootstrap, JavaScript)
Divanshi Pandey	Back-end Development & Database, Documentation (PHP, MySQL)

5. Conclusion

The Timetable & Study Planner Generator is a practical, student-friendly project that automates a task nearly every student deals with every week. By combining a simple, fixed scheduling rule with an interactive drag-and-drop editor, it strikes a balance between automation and the student's own control over their study plan.

From an academic point of view, the project covers the full range of a small web application - front-end design, back-end logic, and database management - using HTML, CSS, Bootstrap, JavaScript, PHP, and MySQL. It is a well-rounded project that is simple to build, easy to demonstrate, and genuinely useful for day-to-day student life.

In conclusion, the Timetable & Study Planner Generator is more than just an academic submission - it is a small step towards making everyday study planning easier and more organised for students.